

MATHS PAPER 2

STUDENT STUDY GUIDE

Everything explained in plain language · Based on past papers

Meet Nova — Your Maths Study Mentor

Nova comes from the Latin for "new," like a star that suddenly shines bright. That's exactly his vibe — turning confusion into those "ohhh" moments. Maths becomes patterns instead of panic. Nova shows you there's always a method behind the madness. When you're stuck, he finds a new way in.

WHAT'S INSIDE

- P1** What the Paper Looks Like
- P2** The 5 Topics That Matter Most
- P3** Step-by-Step Methods with Worked Examples
- P4** Mistakes to Avoid
- P5** Your 14-Day Study Plan

PART 1**What Does the Paper Look Like?**

Paper 2 has **9 questions** that appear in the **same order every year**. This paper is highly predictable — the same topics come up in the same positions. There are no surprises if you prepare correctly.

Q	What it Tests	Marks	Difficulty
Q 1	Statistics: mean, standard deviation, regression line	8–14	Easy
Q 2	Grouped data: drawing ogive, finding median	8–11	Easy
Q 3	Analytical geometry: lines, gradients, area	18–22	Medium
Q 4	Circle geometry using coordinates	18–22	Hard
Q 5	Trig identities and solving equations	18–28	Hard
Q 6	Reading and sketching trig graphs	10–13	Medium
Q 7	3D trig problems (buildings, ramps, angles)	8–11	Medium
Q 8	Euclidean geometry proofs using circle theorems	11–16	Hard
Q 9	Similar triangles and ratio proofs	15–20	Hard

■ **Tip:** 150 marks in 3 hours = about 1 minute per mark. Do not spend 20 minutes stuck on one part. Move on, come back later.

PART 2

The 5 Topics That Give You the Most Marks

These 5 topics appear every single year and make up most of the paper. Master them and you can access **75+ marks out of 150**.

1 Coordinate Geometry

~~20 marks

Q3 & Q4

- Find the gradient between two points
- Write the equation of a line
- Find where two lines cross (simultaneous equations)
- Calculate the area of a triangle or quadrilateral

2 Trig Identities & Equations

~~20 marks

Q5

- Simplify using $\sin^2 x + \cos^2 x = 1$ and $\tan x = \sin x / \cos x$
- Prove that one side equals the other (work on one side only)
- Solve the equation for x in a given interval
- Write the general solution (add $360n$ or $180n$)

3 Circle Theorems

~~15 marks

Q8 & Q4

- Identify which theorem applies from the diagram
- State each angle with the theorem name as reason
- Build a chain: each statement leads to the next
- End with: 'Therefore ... (proved)'

4 Trig Graphs

~~12 marks

Q6

- Read the amplitude: $(\max - \min) \div 2$
- Read the period: x -distance for one complete cycle
- Identify any horizontal or vertical shifts
- Find x -values where graphs intersect, within the domain

5 Statistics & Regression

~~12 marks

Q1 & Q2

- Calculate mean and standard deviation (STAT mode)
- Find the regression line: $y = a + bx$
- Use the equation to predict a value and explain it
- Read the median from an ogive at the 50% mark

PART 3

How to Solve Each Question Type

Follow these steps exactly. They work every time. Each method includes worked examples at three levels of difficulty.

METHOD 1

Coordinate Geometry

Step 1 — Gradient	Formula: $m = (y_2 - y_1) / (x_2 - x_1)$. Plug in both points. Watch the minus signs carefully.
Step 2 — Line equation	Formula: $y - y_1 = m(x - x_1)$. Put in your m and one point. Simplify to $y = mx + c$.
Step 3 — Intersection	Set the two equations equal to each other. Solve simultaneously. Answer is an (x, y) coordinate.
Step 4 — Area	Use the shoelace formula with all vertex coordinates, OR: $\text{Area} = \frac{1}{2} \times \text{base} \times \text{perpendicular height}$.

■ **Tip:** Circle tangent shortcut: The tangent is always perpendicular to the radius. If radius gradient = m , then tangent gradient = $-1/m$.

WORKED EXAMPLE A

Easy

Find the gradient and equation of the line through A(2, 3) and B(6, 11).

Step 1 Find gradient: $m = (11 - 3) / (6 - 2) = 8 / 4 = 2$

Step 2 Use point-slope with A(2, 3): $y - 3 = 2(x - 2)$

Step 3 Expand: $y - 3 = 2x - 4$

Step 4 Simplify: $y = 2x - 1$

✓ **Answer: $m = 2$, Equation: $y = 2x - 1$**

WORKED EXAMPLE B

Standard

Points P(1, 5), Q(4, -1) and R(7, 3) form a triangle. Find the area of triangle PQR.

Step 1 Shoelace formula: $\text{Area} = \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

Step 2 Substitute: $\text{Area} = \frac{1}{2}|1(-1 - 3) + 4(3 - 5) + 7(5 - (-1))|$

Step 3 Calculate: $= \frac{1}{2}|1(-4) + 4(-2) + 7(6)|$

Step 4 $= \frac{1}{2}|-4 - 8 + 42| = \frac{1}{2}|30| = 15$

✓ **Answer: Area = 15 square units**

WORKED EXAMPLE C

Hard

Circle has centre $M(3, -1)$. It intersects at $T(1, 7)$. Find the equation of the tangent at T .

Step 1 Gradient of radius MT : $m_r = (7 - (-1)) / (1 - 3) = 8 / (-2) = -4$

Step 2 Tangent $\perp$ radius, so: $m_t = -1 / (-4) = 1/4$

Step 3 Use point $T(1, 7)$: $y - 7 = \frac{1}{4}(x - 1)$

Step 4 Multiply by 4: $4y - 28 = x - 1$

Step 5 Simplify: $x - 4y + 27 = 0$

✓ **Answer:** Tangent: $y = \frac{1}{4}x + 27/4$ (or $x - 4y + 27 = 0$)

Try It Yourself

Find the gradient and equation of the line through $C(-1, 4)$ and $D(3, -4)$.

Answer: $m = -2$; Equation: $y = -2x + 2$

METHOD 2

Trig Identities and Equations

Step 1 — Spot the pattern

$$\sin^2 x + \cos^2 x = 1 \mid \tan x = \sin x / \cos x \mid 1 - \sin^2 x = \cos^2 x$$

Step 2 — Simplify one side

Work on the more complicated side only. Never move terms across the equals sign when proving.

Step 3 — Solve

Get one trig function alone. Use calculator for the reference angle. Apply the correct quadrant rule.

Step 4 — All solutions

Check the domain. For $\sin/\cos$ add $360^\circ n$. For $\tan$ add $180^\circ n$. List every angle in range.

■ **Warning:** Always check your calculator is in DEGREES mode before any trig calculation.

WORKED EXAMPLE A

Easy

Simplify: $(\sin x \cdot \cos x) / \tan x$

Step 1 Replace $\tan x$ with $\sin x / \cos x$

Step 2 $(\sin x \cdot \cos x) \div (\sin x / \cos x)$

Step 3 $= \sin x \cdot \cos x \cdot (\cos x / \sin x)$

Step 4 $= \cos^2 x$

✓ **Answer:** $\cos^2 x$

WORKED EXAMPLE B

Prove that: $(1 - \cos^2 x) / \sin x = \sin x$

Step 1 Work on the LEFT side only.

Step 2 Use identity: $1 - \cos^2 x = \sin^2 x$

Step 3 So: $\sin^2 x / \sin x$

Step 4 $= \sin x \checkmark$ (matches right side)

✓ Answer: LHS = sin x = RHS ∴ Proved

WORKED EXAMPLE C

Solve for $x \in [-180^\circ; 180^\circ]$: $2\sin^2 x - \sin x - 1 = 0$

Step 1 Factor like a quadratic: let $k = \sin x \rightarrow 2k^2 - k - 1 = 0$

Step 2 Factor: $(2k + 1)(k - 1) = 0$

Step 3 So: $\sin x = -\frac{1}{2}$ OR $\sin x = 1$

Step 4 For $\sin x = 1$: $x = 90^\circ$

Step 5 For $\sin x = -\frac{1}{2}$: reference angle = 30° . Sin is negative in Q3 and Q4.

Step 6 Q3: $x = -(180^\circ - 30^\circ) = -150^\circ$ | Q4: $x = -30^\circ$

Step 7 Check all values are in $[-180^\circ; 180^\circ]$: $\checkmark$

✓ Answer: $x = -150^\circ, x = -30^\circ, x = 90^\circ$

— Try It Yourself

Solve for $x \in [0^\circ; 360^\circ]$: $2\cos x - \sqrt{3} = 0$

Answer: $\cos x = \sqrt{3}/2$; reference angle = 30° ; $x = 30^\circ$ or $x = 330^\circ$

METHOD 3

Circle Theorem Proofs

Step 1 — Study the diagram

Mark every angle you know. Identify tangents, chords, cyclic quads. What must you prove?

Step 2 — State with reason

Write: "angle ABC = angle DEF (theorem name)". The reason in brackets is compulsory.

Step 3 — Build the chain

Each statement leads to the next. Work step by step from known to what you need to prove.

Step 4 — Conclusion

Always end with: 'Therefore [what was proved]'. Never leave it implied.

Theorems you must know word-for-word:

Theorem	What it says
Angle at centre	Angle at centre = $2 \times$ angle at circumference (same arc)
Same segment	Angles in the same segment are equal
Cyclic quadrilateral	Opposite angles of a cyclic quad add up to 180°
Tangent-radius	The tangent is perpendicular to the radius at the point of contact
Equal tangents	Tangents drawn from the same external point are equal in length
Tan-chord angle	Angle between tangent and chord = angle in the alternate segment

WORKED EXAMPLE A

Easy

O is the centre. Angle at centre $AOB = 80^\circ$. Find angle ACB at the circumference on the major arc.

Step 1 Identify: angle at centre vs angle at circumference.

Step 2 Theorem: angle at centre = $2 \times$ angle at circumference (same arc).

Step 3 Angle $ACB = 80^\circ / 2 = 40^\circ$

✓ **Answer: Angle $ACB = 40^\circ$ (Angle at centre = $2 \times$ angle at circumference)**

WORKED EXAMPLE B

Standard

ABCD is a cyclic quadrilateral. Angle $DAB = 115^\circ$. Find angle BCD.

Step 1 ABCD is a cyclic quadrilateral.

Step 2 Opposite angles of a cyclic quad are supplementary.

Step 3 Angle $DAB +$ angle $BCD = 180^\circ$ (opp. angles, cyclic quad)

Step 4 $115^\circ +$ angle $BCD = 180^\circ$

Step 5 Angle $BCD = 65^\circ$

✓ **Answer: Angle $BCD = 65^\circ$**

WORKED EXAMPLE C

Hard

PA and PB are tangents from external point P. Angle $APB = 50^\circ$. Prove triangle APB is isosceles and find angle PAB.

Step 1 $PA = PB$ (tangents from the same external point are equal)

Step 2 Since $PA = PB$, triangle APB is isosceles. $\therefore$ Proved.

Step 3 Angle $PAB =$ angle PBA (base angles of isosceles triangle)

Step 4 Angle PAB + angle PBA + $50^\circ = 180^\circ$ (angles of a triangle)

Step 5 $2 \times \text{angle PAB} = 130^\circ \rightarrow \text{angle PAB} = 65^\circ$

✓ **Answer:** Triangle APB is isosceles (equal tangents); Angle PAB = 65°

— Try It Yourself

O is the centre. Chord AB subtends 70° at the centre. Find the angle AB subtends at the circumference on the major arc.

Answer: Angle at circumference = $70^\circ \div 2 = 35^\circ$ (angle at centre theorem)

METHOD 4

Trig Graphs

Step 1 — Amplitude	Amplitude = (max value – min value) $\div$ 2. Example: max = 3, min = -3 , amplitude = 3.
Step 2 — Period	Period = x-distance for one full cycle. Standard sin/cos = 360° . Repeats in $180^\circ \rightarrow$ period halved.
Step 3 — Shifts	Horizontal shift: graph moved left or right. Vertical shift: check the midline of the graph.
Step 4 — Read solutions	Where do graphs intersect? Read x-values at crossing points. Check every answer is inside the given domain.

■ **Tip:** Equation format: $y = a \cdot \sin(bx + c) + d$ where a = amplitude, period = $360^\circ/b$, c = horizontal shift, d = vertical shift.

WORKED EXAMPLE A

Easy

A graph has maximum 4, minimum -4 , one cycle from 0° to 360° . Write the equation.

Step 1 Amplitude: $a = (4 - (-4)) / 2 = 4$

Step 2 Period = 360° , so $b = 1$

Step 3 No shift visible.

✓ **Answer:** $y = 4 \sin(x)$

WORKED EXAMPLE B

Standard

Function: $y = 2 \sin(x) - 1$, $x \in [0^\circ; 360^\circ]$. State: (a) amplitude, (b) range, (c) period.

Step 1 Amplitude: coefficient of sin = 2, so amplitude = 2.

Step 2 Graph shifts down 1. Max = $2(1) - 1 = 1$. Min = $2(-1) - 1 = -3$.

Step 3 No change to b , so period = 360° .

✓ **Answer: Amplitude = 2; Range: $-3 \leq y \leq 1$; Period = 360°**

WORKED EXAMPLE C

Hard

$f(x) = a \cdot \cos(bx) + d$ passes through $(0^\circ, 5)$, minimum -1 , period 180° . Find a , b and d .

Step 1 Amplitude: $a = (\max - \min) / 2 = (5 - (-1)) / 2 = 3$

Step 2 Vertical shift: $d = (\max + \min) / 2 = (5 + (-1)) / 2 = 2$

Step 3 Period = $360^\circ / b = 180^\circ \rightarrow b = 2$

Step 4 Verify: $f(0) = 3 \cdot \cos(0) + 2 = 3 + 2 = 5$ ✓

✓ **Answer: $a = 3$, $b = 2$, $d = 2 \rightarrow f(x) = 3\cos(2x) + 2$**

→ Try It Yourself

A graph has maximum 3, minimum -1 , one cycle over 360° . Write its equation in the form $y = a \cdot \sin(x) + d$.

Answer: $a = (3 - (-1))/2 = 2$; $d = (3 + (-1))/2 = 1$; Equation: $y = 2\sin(x) + 1$

METHOD 5

Statistics and Regression

Step 1 — Enter data

Go to STAT mode → Edit. Type all values carefully. Check every entry before continuing.

Step 2 — Mean & SD

STAT → Calc → 1-Var Stats. Mean shown as $\bar{x}$. Standard deviation as s_x or σ_x .

Step 3 — Regression line

STAT → Calc → LinReg($a+bx$). Write a and b to at least 2 decimal places.

Step 4 — Predict

Substitute the x -value into $y = a + bx$. Calculate y . Write one sentence explaining the value in real life.

Step 5 — Read the ogive

For median: go to 50% on y -axis. Draw horizontal to curve, then vertical down to x -axis.

■ **Warning:** Most marks lost here come from calculator mistakes, not from not knowing the work. Enter data slowly and check it twice.

WORKED EXAMPLE A

Easy

Dataset: 3, 7, 8, 5, 6. Calculate the mean.

Step 1 Mean = sum of all values ÷ number of values

Step 2 Sum = $3 + 7 + 8 + 5 + 6 = 29$

Step 3 Number of values = 5

Step 4 Mean = $29 \div 5 = 5.8$

✓ **Answer: Mean = 5.8**

WORKED EXAMPLE B

Standard

Regression line: $y = 12.3 + 6.8x$ (study hours vs test score). Predict score for 4 hours study.

Step 1 Substitute $x = 4$: $y = 12.3 + 6.8(4)$

Step 2 $y = 12.3 + 27.2 = 39.5$

Step 3 Interpretation: A student who studies 4 hours is predicted to score approximately 39.5%.

✓ **Answer: $y = 39.5$ — predicted score for 4 hours study**

WORKED EXAMPLE C

Hard

An ogive shows cumulative frequency for 80 learners. Describe how to read the median and IQR.

Step 1 Median: go to 50% of total = 40 on y-axis. Horizontal to curve, vertical down. Read that x-value.

Step 2 Q1: go to 25% = 20 on y-axis. Same process.

Step 3 Q3: go to 75% = 60 on y-axis. Same process.

Step 4 $IQR = Q3 - Q1$

✓ **Answer: Median at 40th value; Q1 at 20th; Q3 at 60th; $IQR = Q3 - Q1$**

— Try It Yourself

Regression equation: $y = 5.2 + 3.1x$. Predict when $x = 10$ and interpret.

Answer: $y = 5.2 + 3.1(10) = 36.2$ — the model predicts a value of approximately 36.2 when $x = 10$.

PART 4

Mistakes That Cost Students Marks

These are the most common reasons marks are dropped — even by students who know the work.

1. Not Finishing the Answer

✗ Students stop at part (a) and miss the easier marks in parts (b) and (c).

✓ Read the whole question before you start. Know where you are going before you write anything.

2. No Reason in Geometry Proofs

✗ Writing 'angle A = angle B' earns 0 marks without a theorem name.

✓ After every angle you state, ask yourself: WHY is that true? Write the theorem name in brackets.

3. Calculator in Wrong Mode

✗ If your calculator is in Radians instead of Degrees, every trig answer will be wrong — and the numbers look plausible, making it worse.

✓ Before every trig question: press Mode and confirm it says DEG.

4. Sign Errors in Algebra

✗ Especially common when subtracting negative numbers or substituting negative coordinates into formulas.

✓ Write each substitution step separately. Never skip lines. Check: does the sign make logical sense?

5. Wrong Solutions for Trig Domain

✗ Giving answers outside the specified range, or missing valid answers inside it.

✓ Underline the domain before solving. Tick each answer that falls inside the range.

6. Correct Setup, Wrong Arithmetic

✗ The most common mark-loser at every level: perfect setup, calculation error at the last step.

✓ Substitute your answer back into the original equation to verify. Takes 30 seconds, saves marks.

PART 5

Your 14-Day Study Plan

Follow this plan day by day. Topics earlier in the plan are worth more marks. **Do not skip ahead.**

Days 1–3	Coordinate Geometry	~20 marks
→	Open any past Paper 2. Find Q3.	
→	Do 20 gradient calculations. Use formula: $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$ every time.	
→	Do 20 line equations. Use: $y - y_1 = m(x - x_1)$ every time.	
→	Do 10 area questions using coordinates.	
→	Goal: Set up any coordinate geometry question in under 2 minutes.	
Days 4–6	Trig Identities & Equations	~20 marks
→	Write the 3 main identities on a card. Stick it where you study.	
→	Practise simplifying 10 trig expressions per day using those identities.	
→	Practise proving 5 identities per day. Work on one side only.	
→	Practise solving trig equations and writing general solutions.	
→	Goal: Recognise which identity to use within 10 seconds of reading.	
Days 7–8	Trig Graphs	~12 marks
→	Find Q6 from 3 different past papers.	
→	For each graph: write amplitude, period, and any shifts.	
→	Practise writing the equation of a graph from a sketch.	
→	Practise reading x-values where two graphs cross.	
→	Goal: Complete a full Q6 in 10 minutes or less.	
Days 9–10	Circle Theorems	~15 marks
→	Write each theorem on a card with its exact wording. Learn them.	
→	Find Q8 from 3 past papers. For each statement, say the reason aloud.	
→	Practise identifying which theorem applies from the diagram alone.	
→	Work through 2 complete proofs per day.	
→	Goal: State any theorem correctly, word-for-word, without hesitation.	
Days 11–12	Statistics & Regression	~12 marks

- Practise finding mean and standard deviation on your calculator.
- Practise finding the regression equation. Write a and b each time.
- Practise substituting an x-value and interpreting the answer in words.
- Practise reading the median from an ogive (50% on y-axis).
- Goal: Full marks on Q1 and Q2 every time. These should be automatic.

Day 13 Mixed Timed Practice

Test
yourself

- Set a timer. Do Q1 in 12 min. Q3 in 20 min. Q5 in 20 min. Q6 in 12 min.
- Work in silence. Do not check notes.
- After: mark your answers. Write down exactly where you lost marks.
- Redo only the parts you got wrong.

Day 14 Full Past Paper

Exam
simulation

- Print a full past Paper 2 (most recent year available).
- Set a 3-hour timer. Silence your phone. No notes. Exam conditions.
- When done: mark it fully. Write down your total mark.
- List 3 specific things to fix. Revise those before the exam.

EVERY DAY

Daily 5-Minute Warm-Up

Do this before every main study session to keep all skills sharp.

Task	Why
✓ 5 algebra simplifications	Keeps your algebra sharp so you do not make errors in the exam.
✓ 3 trig equations	Builds speed. You want this to feel automatic.
✓ 3 geometry theorem statements	Say them out loud. Hearing yourself say them helps you remember.
✓ 1 graph reading question	Quick and easy. Keeps graph skills fresh.
✓ 1 coordinate problem	Keeps the method in your muscle memory.

You can do this. The paper is predictable. The methods are learnable. Master the 5 core topics, avoid the 6 common mistakes, and follow the 14-day plan. That is all you need to do.

**NATIONAL
SENIOR CERTIFICATE
GRADE 12**

**MATHEMATICS
PAPER 2**

PREDICTIVE EXAMINATION — 2026

MARKS: 150
TIME: 3 hours

This question paper consists of 10 pages and 1 information sheet.

This is a predictive paper based on the recurring structure of NSC Paper 2 (2021-2025). It is intended for revision purposes only and is not an official Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

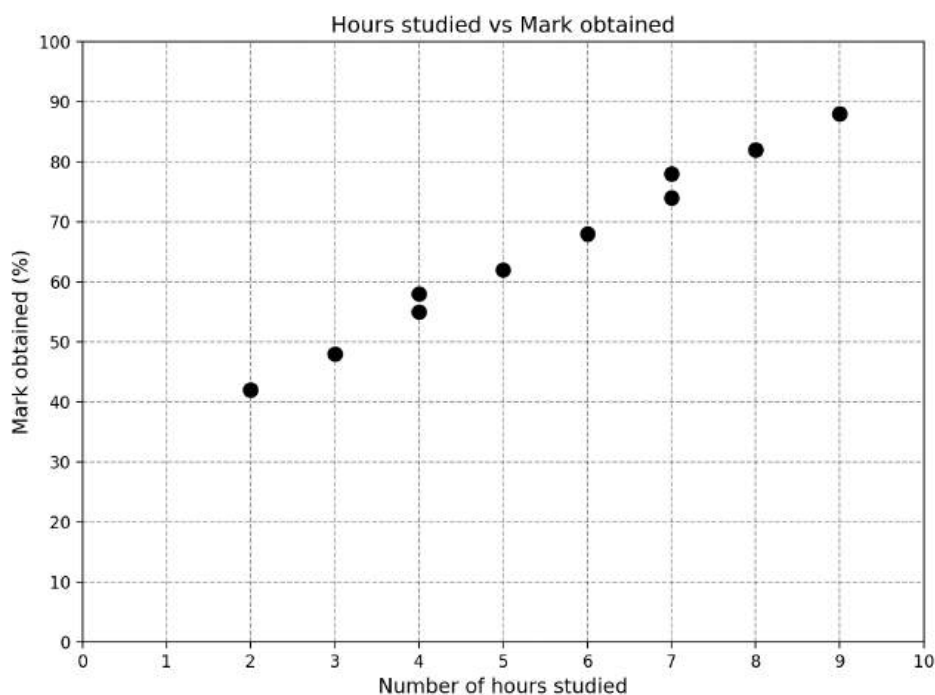
1. This question paper consists of **NINE** questions.
2. Answer **ALL** the questions in the ANSWER BOOK provided.
3. Clearly show **ALL** calculations, diagrams, graphs, et cetera that you have used in determining your answers.
4. Answers only will **NOT** necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to **TWO** decimal places, unless stated otherwise.
7. Diagrams are **NOT** necessarily drawn to scale.
8. An information sheet with formulae is included at the end of the question paper.
9. Number the answers correctly according to the numbering system used in this question paper.
10. Write neatly and legibly.

QUESTION 1

A teacher recorded the number of hours that 10 learners spent studying for a test and the percentage mark each learner obtained. The data is shown in the table below.

Hours studied (x)	2	3	4	4	5	6	7	7	8	9
Mark (%) (y)	42	48	55	58	62	68	74	78	82	88

The scatter plot of the data is shown below.



- 1.1 Calculate the mean of the marks. (2)
- 1.2 Calculate the standard deviation of the marks. (2)
- 1.3 Determine how many learners scored within ONE standard deviation of the mean. (2)
- 1.4 Determine the equation of the least squares regression line in the form $\hat{y} = a + bx$. Round a and b to TWO decimal places. (3)
- 1.5 Use the equation in QUESTION 1.4 to predict the percentage mark of a learner who studies for 7,5 hours. (2)
- 1.6 Comment on the strength and direction of the linear relationship between hours studied and marks obtained. (2)

[13]

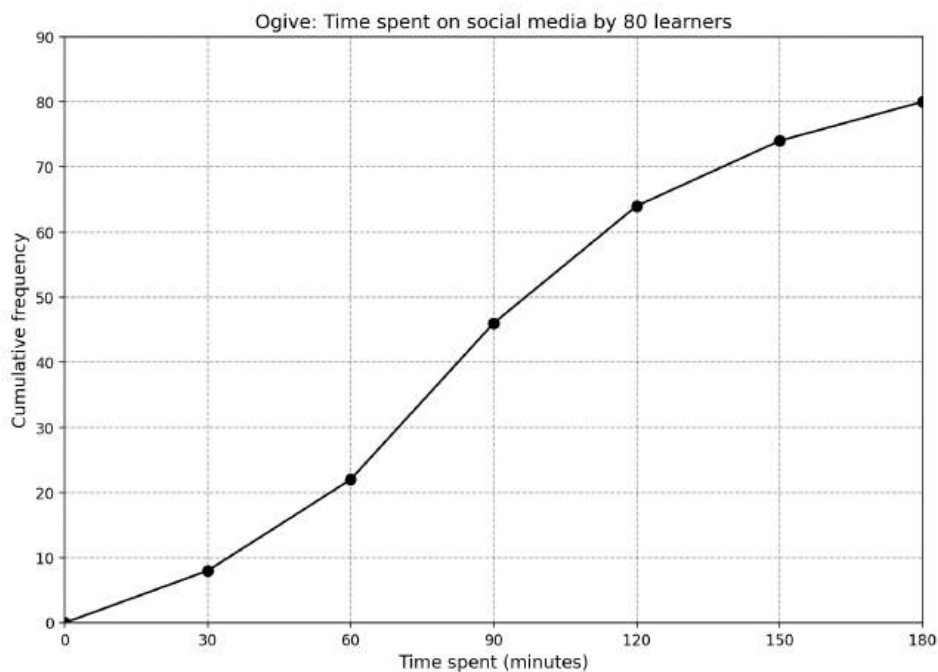
QUESTION 2

A school recorded the time (in minutes) spent on social media in one day by 80 learners. The information is summarised in the frequency table below.

Time (minutes)	Frequency	Cumulative frequency
$0 \leq t < 30$	8	8
$30 \leq t < 60$	14	22
$60 \leq t < 90$	24	46
$90 \leq t < 120$	18	64
$120 \leq t < 150$	10	74
$150 \leq t < 180$	6	80

- 2.1 Identify the modal class. (1)
- 2.2 Calculate the estimated mean time spent on social media. (3)
- 2.3 Draw an ogive (cumulative frequency curve) for the data on the grid provided in the ANSWER BOOK. (3)

An ogive of the data is shown below.

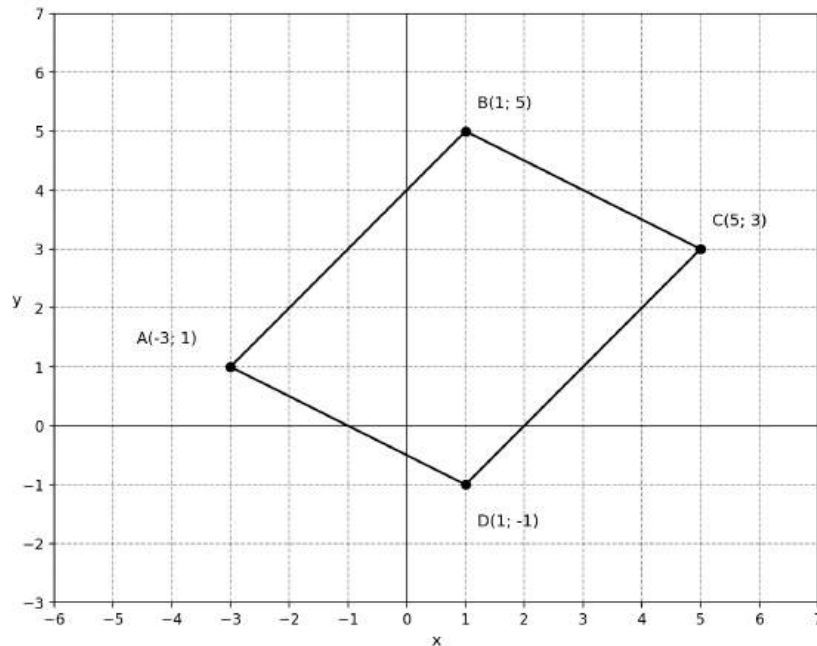


- 2.4 Use the ogive to estimate the median time. (2)
- 2.5 Calculate the percentage of learners who spent more than 90 minutes on social media. (2)

[11]

QUESTION 3

In the diagram, ABCD is a quadrilateral with vertices $A(-3 ; 1)$, $B(1 ; 5)$, $C(5 ; 3)$ and $D(1 ; -1)$ in a Cartesian plane.

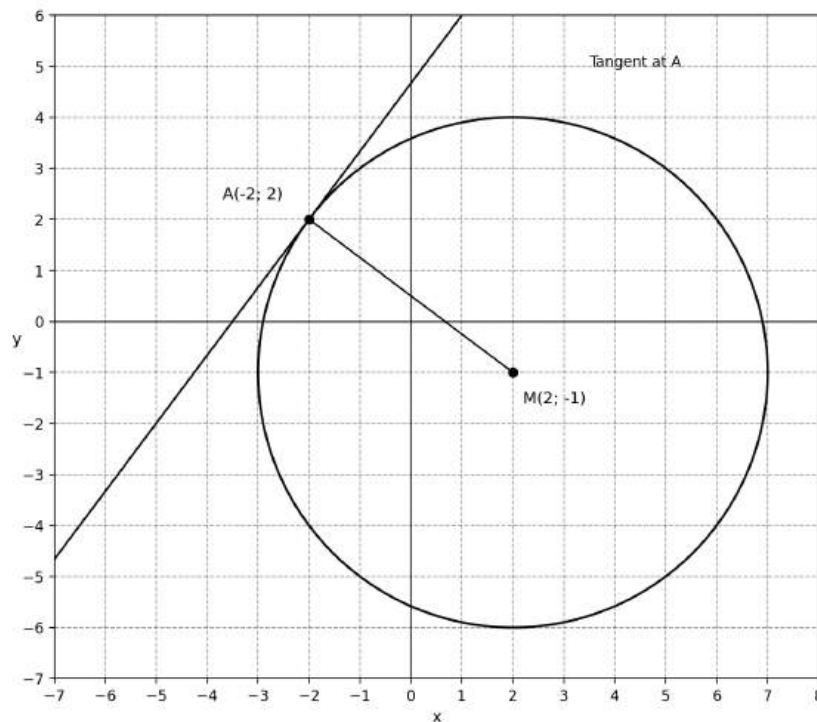


- 3.1 Calculate the gradient of AB. (2)
- 3.2 Determine the equation of line AB in the form $y = mx + c$. (3)
- 3.3 Calculate the length of BC. Leave your answer in surd form. (2)
- 3.4 Determine the coordinates of M, the midpoint of AC. (2)
- 3.5 Show that ABCD is a parallelogram. (4)
- 3.6 Calculate the size of $\hat{A}$, the angle of the parallelogram at vertex A. Round to ONE decimal place. (4)
- 3.7 Calculate the area of parallelogram ABCD. (3)

[20]

QUESTION 4

In the diagram, a circle has centre $M(2; -1)$ and passes through the point $A(-2; 2)$. The line drawn through A is a tangent to the circle at A .



- 4.1 Determine the length of the radius MA . (2)
- 4.2 Write down the equation of the circle in the form $(x - a)^2 + (y - b)^2 = r^2$. (2)
- 4.3 Calculate the gradient of MA . (2)
- 4.4 Determine the equation of the tangent to the circle at A in the form $y = mx + c$. (4)
- 4.5 K is the point diametrically opposite to A on the circle. Calculate the coordinates of K . (3)
- 4.6 $P(7; 6)$ is a point outside the circle. Calculate the length of the tangent drawn from P to the circle. (3)
- 4.7 Determine, with reasons, whether the point $N(5; 3)$ lies inside, on or outside the circle. (3)

[19]

QUESTION 5

5.1 Simplify the following expression to a single trigonometric ratio:

$$\frac{\sin(180^\circ + x) \cdot \cos(360^\circ - x) \cdot \tan(180^\circ - x)}{\cos(90^\circ - x) \cdot \sin(-x)} \quad (5)$$

5.2 Prove the identity:

$$(1 - \cos 2x) / \sin 2x = \tan x \quad (4)$$

5.3 Determine the general solution of:

$$\cos(2x - 30^\circ) = \frac{1}{2} \quad (5)$$

5.4 Solve for $x \in [0^\circ ; 360^\circ]$:

$$2 \sin^2 x + 3 \cos x - 3 = 0 \quad (6)$$

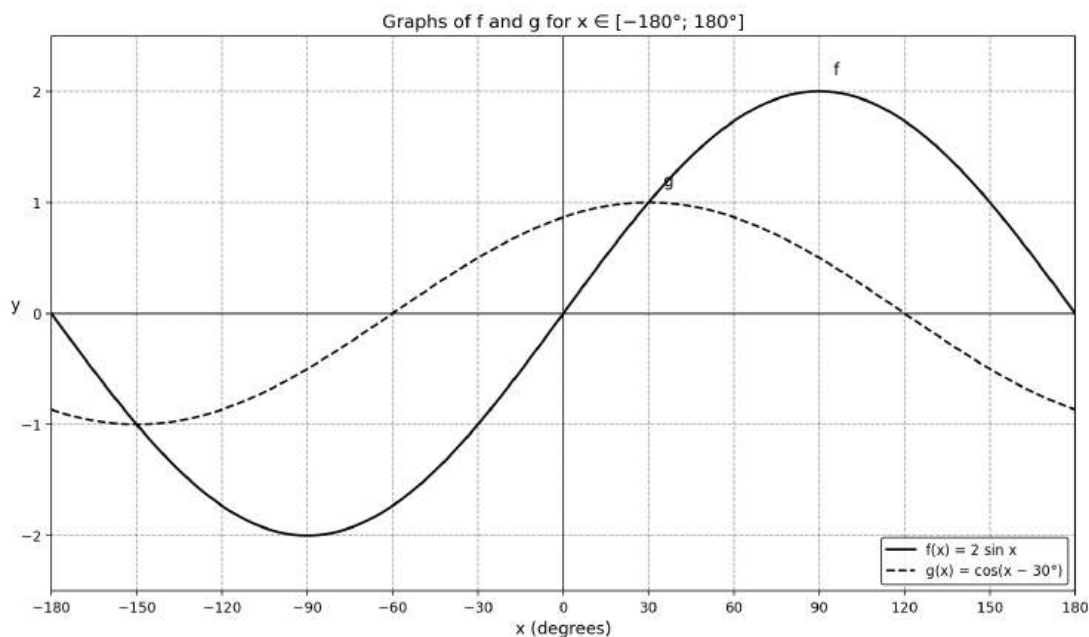
5.5 Without using a calculator, evaluate:

$$\sin 75^\circ \cdot \cos 15^\circ - \cos 75^\circ \cdot \sin 15^\circ \quad (2)$$

[22]

QUESTION 6

In the diagram, the graphs of $f(x) = 2 \sin x$ and $g(x) = \cos(x - 30^\circ)$ are drawn for $x \in [-180^\circ; 180^\circ]$.



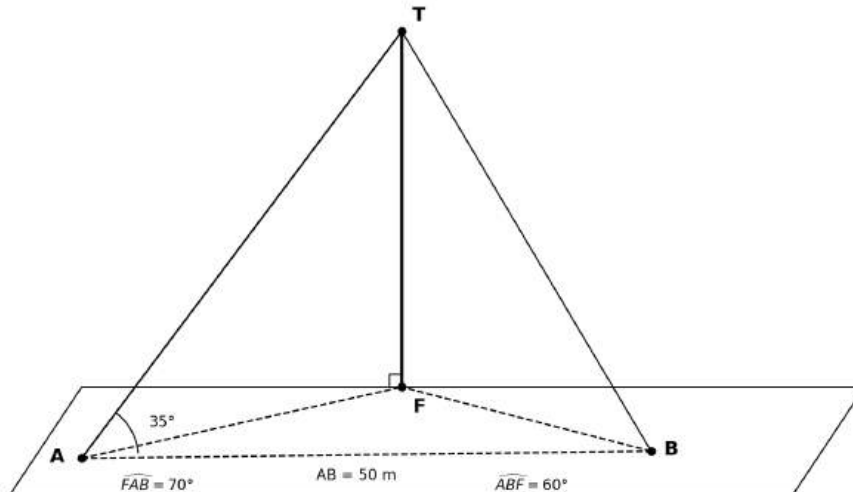
- 6.1 Write down the amplitude of f . **(1)**
- 6.2 Write down the period of g . **(1)**
- 6.3 Determine the values of x in the interval $[-180^\circ; 180^\circ]$ for which $f(x) = g(x)$. **(5)**
- 6.4 Hence, or otherwise, determine the values of $x \in [-180^\circ; 180^\circ]$ for which $f(x) > g(x)$. **(3)**
- 6.5 Write down the maximum value of $h(x) = f(x) - 1$. **(1)**
- 6.6 If the graph of g is shifted 30° to the LEFT to form graph k , write down the equation of k in its simplest form. **(2)**

[13]

QUESTION 7

TF is a vertical tower standing on horizontal ground. The points A and B are on the ground, at the same level as F, the foot of the tower. T is the top of the tower. The angle of elevation from A to T is 35° . $AB = 50$ m, $\widehat{FAB} = 70^\circ$ and $\widehat{ABF} = 60^\circ$.

Vertical tower TF with observation points A and B on horizontal ground

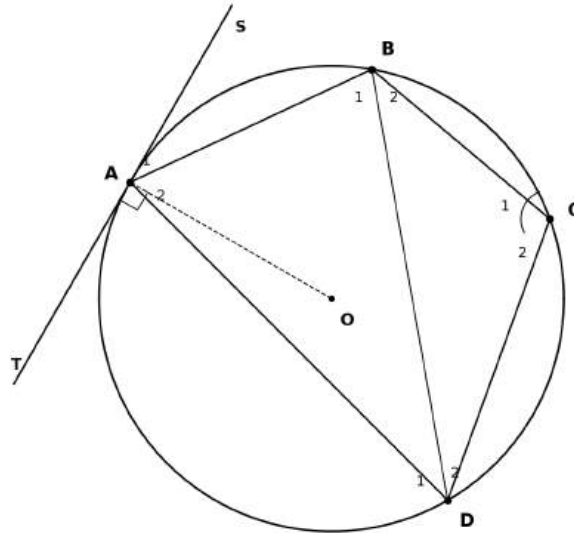


- 7.1 Write down the size of $\angle AFB$. (1)
- 7.2 Show that $AF = (50 \sin 60^\circ) / \sin 50^\circ$. (3)
- 7.3 Hence, calculate the height of the tower TF, correct to TWO decimal places. (3)
- 7.4 Calculate the area of triangle AFB. (4)

[11]

QUESTION 8

In the diagram below, O is the centre of the circle. A , B , C and D are points on the circle. ST is a tangent to the circle at A . BD is drawn. $\angle TAB = 55^\circ$ and $\angle ABD = 30^\circ$.



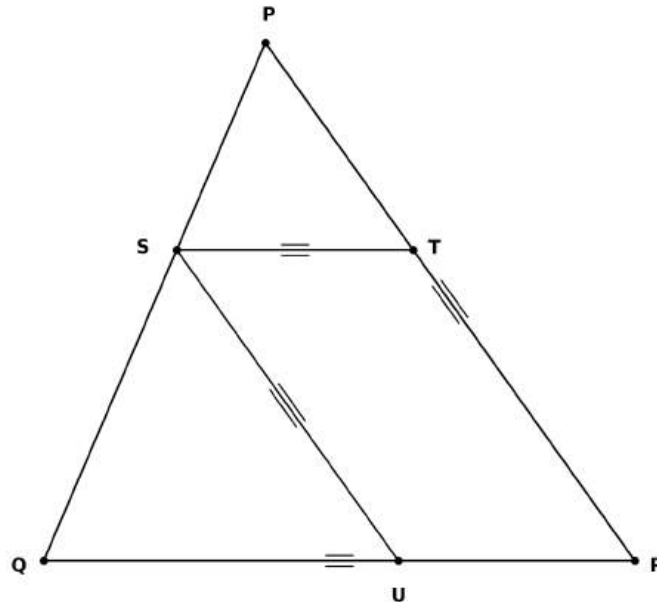
Determine, giving reasons, the size of:

- | | |
|--|------------|
| 8.1 $\angle ADB$ | (2) |
| 8.2 $\angle AOB$ (the angle subtended at the centre by chord AB on the major-arc side) | (2) |
| 8.3 $\angle BAD$ | (3) |
| 8.4 $\angle BCD$ | (2) |
| 8.5 $\angle AOD$ | (2) |
| 8.6 Show that $\angle SAD = \angle ABD$, giving a reason for your statement. | (2) |

[13]

QUESTION 9

In the diagram, $\triangle PQR$ is drawn. S is a point on PQ and T is a point on PR such that $ST \parallel QR$. U is a point on QR such that $SU \parallel PR$. It is given that $PS : SQ = 2 : 3$.



- 9.1 Prove that STRU is a parallelogram. (4)
- 9.2 Determine, with reasons, the value of the ratio $ST : QR$. (3)
- 9.3 Determine, with reasons, the value of the ratio $SU : PR$. (3)
- 9.4 Prove that $\triangle PST \sim \triangle PQR$. (4)
- 9.5 Hence, determine the ratio $area \triangle PST : area \triangle PQR$. (4)

[18]

TOTAL: 150

INFORMATION SHEET: MATHEMATICS

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

$$A = P(1 + ni) \quad A = P(1 - ni) \quad A = P(1 - i)^n \quad A = P(1 + i)^n$$

$$T_n = a + (n - 1)d \quad S_n = (n/2)[2a + (n - 1)d]$$

$$T_n = a r^{n-1} \quad S_n = a(r^n - 1)/(r - 1); r \neq 1$$

$$S_\infty = a / (1 - r); -1 < r < 1$$

$$F = x[(1 + i)^n - 1] / i \quad P = x[1 - (1 + i)^{-n}] / i$$

$$f'(x) = \lim_{h \rightarrow 0} [f(x + h) - f(x)] / h$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M(x; y) = ((x_1 + x_2)/2; (y_1 + y_2)/2)$$

$$y = mx + c \quad y - y_1 = m(x - x_1) \quad m = (y_2 - y_1)/(x_2 - x_1) \quad m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: a / \sin A = b / \sin B = c / \sin C$$

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A$$

$$\text{Area } \triangle ABC = \frac{1}{2} \cdot ab \cdot \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \quad \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha = 2 \cos^2 \alpha - 1$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\bar{x} = (\Sigma x) / n \quad \sigma^2 = (\Sigma (x_i - \bar{x})^2) / n$$

$$P(A) = n(A) / n(S) \quad P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx \quad b = (\Sigma (x - \bar{x})(y - \bar{y})) / (\Sigma (x - \bar{x})^2)$$

**NATIONAL
SENIOR CERTIFICATE
GRADE 12**

**MATHEMATICS
PAPER 2**

MARKING GUIDELINE

PREDICTIVE EXAMINATION — 2026

MARKS: 150

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy (CA) applies in ALL aspects of the marking guideline.
- Penalise the candidate for using the wrong formula only ONCE per question.
- Method marks (M) are awarded for using the correct method.
- Accuracy marks (A) are awarded for the correct numerical answer.
- Reason marks (R) are awarded for stating the correct theorem or reason.
- Where alternative methods are used, the candidate is awarded the appropriate marks.

This memorandum accompanies a predictive paper based on the recurring structure of NSC Paper 2 (2021-2025). It is intended for revision purposes only and is not an official Department of Basic Education marking guideline.

QUESTION 1

1.1 Mean of the marks (2 marks)

$$\begin{aligned}\text{Mean} &= (\Sigma x) \div n = (42 + 48 + 55 + 58 + 62 + 68 + 74 + 78 + 82 + 88) \div 10 \checkmark(M) \\ &= 655 \div 10\end{aligned}$$

$$\text{Mean} = 65,5\% \checkmark(A)$$

1.2 Standard deviation of the marks (2 marks)

Using calculator (STAT mode, 1-Var Stats): $\checkmark(M)$

$$\sigma \approx 14,32$$

$$\sigma \approx 14,32 \checkmark(A)$$

(Population standard deviation. Accept σ between 14,30 and 14,33.)

1.3 Number of learners within ONE SD of the mean (2 marks)

$$\begin{aligned}\text{Interval: } (\bar{x} - \sigma ; \bar{x} + \sigma) &= (65,5 - 14,32 ; 65,5 + 14,32) \checkmark(M) \\ &= (51,18 ; 79,82)\end{aligned}$$

Marks within this interval: **55, 58, 62, 68, 74, 78**

6 learners scored within ONE standard deviation of the mean.
 $\checkmark(A)$

1.4 Equation of the least-squares regression line (3 marks)

Using calculator: STAT $\rightarrow$ CALC $\rightarrow$ LinReg(a + bx) $\checkmark(M)$

$$a \approx 29,13 \checkmark(A)$$

$$b \approx 6,61 \checkmark(A)$$

$$\hat{y} = 29,13 + 6,61x$$

1.5 Predicted mark for 7,5 hours of study (2 marks)

$$\begin{aligned}\hat{y} &= 29,13 + 6,61(7,5) \checkmark(M) \\ &= 29,13 + 49,575 \\ &= 78,705\end{aligned}$$

$$\text{Predicted mark} \approx 78,71\% \checkmark(CA)$$

(Accept any answer in the range 78,50% – 79,00% depending on rounding of a and b.)

1.6 Strength and direction of the relationship (2 marks)

The points lie very close to a straight line that rises from left to right. $\checkmark(\checkmark)$

There is a **strong positive** linear relationship between hours studied and marks obtained. $\checkmark(\checkmark)$

(The correlation coefficient $r \approx 0,996$, which is very close to 1, confirming a strong positive correlation.)

[13]

QUESTION 2

2.1 Modal class (1 mark)

$60 \leq t < 90$ minutes ✓ (A) (highest frequency = 24)

2.2 Estimated mean time spent on social media (3 marks)

Use class midpoints:

Midpoints: 15 ; 45 ; 75 ; 105 ; 135 ; 165 ✓ (M)

$$\begin{aligned}\Sigma fx &= 8(15) + 14(45) + 24(75) + 18(105) + 10(135) + 6(165) \checkmark (M) \\ &= 120 + 630 + 1800 + 1890 + 1350 + 990 \\ &= 6\,780\end{aligned}$$

$$\text{Mean} = \Sigma fx \div \Sigma f = 6\,780 \div 80$$

Estimated mean = 84,75 minutes ✓ (A)

2.3 Drawing the ogive (3 marks)

The ogive is plotted using the upper boundary of each class against the cumulative frequency:

Upper boundary (min)	0	30	60	90	120	150	180
Cumulative frequency	0	8	22	46	64	74	80

Marks: ✓ Starting point at (0 ; 0) ✓ (M)

✓ All seven points correctly plotted ✓ (A)

✓ Smooth ogive shape ending at (180 ; 80) ✓ (A)

2.4 Median time (2 marks)

On the ogive, locate cumulative frequency = 40 (since $n \div 2 = 80 \div 2 = 40$). ✓ (M)

Read horizontally to the curve, then vertically down to the time axis.

Estimated median $\approx$ 82 minutes ✓ (A)

(Accept any answer between 80 and 84 minutes when read from a hand-drawn ogive. Algebraic interpolation gives exactly 82,5 minutes.)

2.5 Percentage of learners who spent more than 90 minutes (2 marks)

Number of learners spending more than 90 minutes = $80 - 46 = 34$ ✓ (M)

Percentage = $(34 \div 80) \times 100\%$

Percentage = 42,5% ✓ (A)

[11]

QUESTION 3

3.1 Gradient of AB (2 marks)

A(-3 ; 1), B(1 ; 5)

$$m_{AB} = (y_2 - y_1) / (x_2 - x_1) = (5 - 1) / (1 - (-3)) \checkmark (M)$$

$$= 4 / 4$$

$$m_{AB} = 1 \checkmark (A)$$

3.2 Equation of line AB in the form $y = mx + c$ (3 marks)

Using point A(-3 ; 1) and $m = 1$:

$$y - y_1 = m(x - x_1) \checkmark (M)$$

$$y - 1 = 1(x - (-3))$$

$$y - 1 = x + 3 \checkmark (A)$$

$$y = x + 4 \checkmark (A)$$

3.3 Length of BC (2 marks)

B(1 ; 5), C(5 ; 3)

$$BC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(5 - 1)^2 + (3 - 5)^2} \checkmark (M)$$

$$= \sqrt{16 + 4} = \sqrt{20}$$

$$BC = 2\sqrt{5} \text{ units } \checkmark (A)$$

3.4 Coordinates of M, midpoint of AC (2 marks)

A(-3 ; 1), C(5 ; 3)

$$M = ((x_1 + x_2)/2 ; (y_1 + y_2)/2) \checkmark (M)$$

$$= ((-3 + 5)/2 ; (1 + 3)/2)$$

$$M(1 ; 2) \checkmark (A)$$

3.5 Show that ABCD is a parallelogram (4 marks)

Method: show that the diagonals bisect each other.

$$\text{Midpoint of AC: } M = ((-3 + 5)/2 ; (1 + 3)/2) = (1 ; 2) \checkmark (M)$$

$$\text{Midpoint of BD: } ((1 + 1)/2 ; (5 + (-1))/2) = (1 ; 2) \checkmark (A)$$

$$\text{Midpoint of AC} = \text{Midpoint of BD} \checkmark (R)$$

$\therefore$ The diagonals bisect each other.

$$\therefore \text{ABCD is a parallelogram (diagonals bisect each other). } \checkmark (R)$$

(Alternative: show $m_{AB} = m_{DC} = 1$ AND $m_{BC} = m_{AD} = -\frac{1}{2}$, i.e. both pairs of opposite sides are parallel.)

3.6 Size of angle A in the parallelogram (4 marks)

$\hat{A}$ is the interior angle between sides AB and AD.

$$m_{AB} = 1; m_{AD} = (1 - (-1)) / (-3 - 1) = 2 / (-4) = -\frac{1}{2} \checkmark (M)$$

$$\text{Angle of inclination of AB: } \tan \theta_1 = 1 \rightarrow \theta_1 = 45,0^\circ$$

$$\text{Angle of inclination of AD: } \tan \theta_2 = -\frac{1}{2} \rightarrow \theta_2 = 180^\circ - 26,565^\circ = 153,435^\circ \checkmark (A)$$

$$\hat{A} = 153,435^\circ - 45,0^\circ = 108,435^\circ \checkmark (M)$$

$$\hat{A} \approx 108,4^\circ \checkmark (A)$$

(Alternative method using vectors / cosine rule on triangle ABD also accepted.)

3.7 Area of parallelogram ABCD (3 marks)

Method: Area = base $\times$ perpendicular height, OR use the formula:

$$\text{Area of parallelogram} = |AB| \times |AD| \times \sin A \checkmark (M)$$

$$|AB| = \sqrt{4^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$$

$$|AD| = \sqrt{((-3 - 1)^2 + (1 - (-1))^2)} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5}$$

$$\text{Area} = 4\sqrt{2} \times 2\sqrt{5} \times \sin(108,435^\circ) \checkmark (M)$$

$$= 8\sqrt{10} \times 0,9487 \approx 25,30 \times 0,9487$$

$$\text{Area} \approx 24 \text{ square units} \checkmark (A)$$

(Most efficient: use the shoelace formula directly: $\text{Area} = \frac{1}{2}|x_A(y_B - y_D) + x_B(y_C - y_A) + x_C(y_D - y_B) + x_D(y_A - y_C)| = \frac{1}{2}|(-3)(5 - (-1)) + 1(3 - 1) + 5(-1 - 5) + 1(1 - 3)| = \frac{1}{2}|-18 + 2 - 30 - 2| = \frac{1}{2}(48) = 24.$)

[20]

QUESTION 4

4.1 Length of radius MA (2 marks)

$$M(2 ; -1), A(-2 ; 2)$$

$$\begin{aligned} MA &= \sqrt{(-2 - 2)^2 + (2 - (-1))^2} \checkmark(M) \\ &= \sqrt{(-4)^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} \end{aligned}$$

$$\boxed{MA = 5 \text{ units } \checkmark(A)}$$

4.2 Equation of the circle (2 marks)

$$\text{Standard form: } (x - a)^2 + (y - b)^2 = r^2 \checkmark(M)$$

$$\text{Centre } M(2 ; -1), r = 5:$$

$$\boxed{(x - 2)^2 + (y + 1)^2 = 25 \checkmark(A)}$$

4.3 Gradient of MA (2 marks)

$$\begin{aligned} m_{MA} &= (y_2 - y_1) / (x_2 - x_1) = (2 - (-1)) / (-2 - 2) \checkmark(M) \\ &= 3 / (-4) \end{aligned}$$

$$\boxed{m_{MA} = -3/4 \checkmark(A)}$$

4.4 Equation of the tangent at A (4 marks)

The tangent is perpendicular to the radius at the point of contact.

$$m_{\text{tangent}} \times m_{MA} = -1 \checkmark(R)$$

$$m_{\text{tangent}} = -1 \div (-3/4) = 4/3 \checkmark(A)$$

Using A(-2 ; 2) and $m = 4/3$:

$$y - 2 = (4/3)(x - (-2)) \checkmark(M)$$

$$y - 2 = (4/3)x + 8/3$$

$$y = (4/3)x + 8/3 + 2$$

$$\boxed{y = (4/3)x + 14/3 \checkmark(A)}$$

4.5 Coordinates of K (diametrically opposite to A) (3 marks)

M is the midpoint of diameter AK. $\checkmark(R)$

$$\text{Let } K(x ; y). \text{ Then } ((-2 + x)/2 ; (2 + y)/2) = (2 ; -1) \checkmark(M)$$

$$(-2 + x)/2 = 2 \rightarrow x = 6$$

$$(2 + y)/2 = -1 \rightarrow y = -4$$

$$\boxed{K(6 ; -4) \checkmark(A)}$$

4.6 Length of tangent from P(7 ; 6) to the circle (3 marks)

Let T be the point of tangency. Then $\triangle MTP$ is right-angled at T (radius $\perp$ tangent).

By Pythagoras: $PT^2 = MP^2 - MT^2 \checkmark(R)$

$$MP^2 = (7 - 2)^2 + (6 - (-1))^2 = 25 + 49 = 74 \checkmark(M)$$

$$MT^2 = r^2 = 25$$

$$PT^2 = 74 - 25 = 49$$

PT = 7 units ✓(A)

4.7 Position of N(5 ; 3) relative to the circle (3 marks)

Calculate the distance from M to N:

$$MN^2 = (5 - 2)^2 + (3 - (-1))^2 = 3^2 + 4^2 = 9 + 16 \checkmark(M)$$

$$MN^2 = 25 \checkmark(A)$$

Compare MN^2 with $r^2 = 25$:

$$MN^2 = r^2$$

Since $MN^2 = r^2$, point N lies ON the circle. ✓(R)

(Verify by substitution into the equation: $(5 - 2)^2 + (3 + 1)^2 = 9 + 16 = 25 \checkmark$)

[19]

QUESTION 5

5.1 Simplify the expression (5 marks)

Numerator:

$$\sin(180^\circ + x) = -\sin x \checkmark(R)$$

$$\cos(360^\circ - x) = \cos x \checkmark(R)$$

$$\tan(180^\circ - x) = -\tan x \checkmark(R)$$

$$\begin{aligned}\text{Numerator} &= (-\sin x)(\cos x)(-\tan x) = \sin x \cdot \cos x \cdot \tan x \\ &= \sin x \cdot \cos x \cdot (\sin x / \cos x) = \sin^2 x \checkmark(M)\end{aligned}$$

Denominator:

$$\cos(90^\circ - x) = \sin x ; \sin(-x) = -\sin x$$

$$\text{Denominator} = \sin x \cdot (-\sin x) = -\sin^2 x$$

Combine:

$$\text{Expression} = \sin^2 x \div (-\sin^2 x)$$

$$= -1 \checkmark(A)$$

5.2 Prove the identity $(1 - \cos 2x) / \sin 2x = \tan x$ (4 marks)

$$\text{LHS: } (1 - \cos 2x) / \sin 2x$$

$$\text{Use the identities: } \cos 2x = 1 - 2 \sin^2 x \text{ and } \sin 2x = 2 \sin x \cos x \checkmark(R)$$

$$\text{LHS} = (1 - (1 - 2 \sin^2 x)) / (2 \sin x \cos x) \checkmark(M)$$

$$= 2 \sin^2 x / (2 \sin x \cos x) \checkmark(A)$$

$$= \sin x / \cos x \checkmark(M)$$

$$= \tan x = \text{RHS}$$

$$\therefore (1 - \cos 2x) / \sin 2x = \tan x \text{ (proved)}$$

5.3 General solution of $\cos(2x - 30^\circ) = \frac{1}{2}$ (5 marks)

$$\text{Reference angle: } \cos^{-1}(\frac{1}{2}) = 60^\circ \checkmark(M)$$

$$\text{Cosine is positive in Quadrants I and IV. } \checkmark(R)$$

$$\therefore 2x - 30^\circ = 60^\circ + 360^\circ k \quad \text{OR} \quad 2x - 30^\circ = -60^\circ + 360^\circ k, \quad k \in \mathbb{Z} \checkmark(A)$$

$$\text{Case 1: } 2x = 90^\circ + 360^\circ k \rightarrow x = 45^\circ + 180^\circ k \checkmark(A)$$

$$\text{Case 2: } 2x = -30^\circ + 360^\circ k \rightarrow x = -15^\circ + 180^\circ k \checkmark(A)$$

$$x = 45^\circ + 180^\circ k \quad \text{OR} \quad x = -15^\circ + 180^\circ k, \quad k \in \mathbb{Z}$$

5.4 Solve $2 \sin^2 x + 3 \cos x - 3 = 0$ for $x \in [0^\circ; 360^\circ]$ (6 marks)

$$\text{Replace } \sin^2 x \text{ with } (1 - \cos^2 x): \checkmark(M)$$

$$2(1 - \cos^2 x) + 3 \cos x - 3 = 0$$

$$2 - 2 \cos^2 x + 3 \cos x - 3 = 0$$

$$-2 \cos^2 x + 3 \cos x - 1 = 0$$

$$2 \cos^2 x - 3 \cos x + 1 = 0 \checkmark(A)$$

$$\text{Factorise: } (2 \cos x - 1)(\cos x - 1) = 0 \checkmark(M)$$

$$\therefore \cos x = \frac{1}{2} \quad \text{OR} \quad \cos x = 1$$

$$\cos x = \frac{1}{2}: \text{reference angle} = 60^\circ; \cos \text{ positive in Q I and Q IV}$$

$$x = 60^\circ \quad \text{or} \quad x = 360^\circ - 60^\circ = 300^\circ \quad \checkmark(\text{A})$$

$$\cos x = 1: x = 0^\circ \quad \text{or} \quad x = 360^\circ \quad \checkmark(\text{A})$$

$$x = 0^\circ, \quad 60^\circ, \quad 300^\circ, \quad 360^\circ \quad \checkmark(\text{A})$$

5.5 Evaluate $\sin 75^\circ \cdot \cos 15^\circ - \cos 75^\circ \cdot \sin 15^\circ$ without a calculator (2 marks)

Recognise the compound-angle identity: $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta \quad \checkmark(\text{M})$

$$\therefore \sin 75^\circ \cdot \cos 15^\circ - \cos 75^\circ \cdot \sin 15^\circ = \sin(75^\circ - 15^\circ) = \sin 60^\circ$$

$$= \sqrt{3} / 2 \quad \checkmark(\text{A})$$

[22]

QUESTION 6

6.1 Amplitude of f (1 mark)

$f(x) = 2 \sin x$; amplitude is the coefficient of $\sin x$.

Amplitude = 2 ✓ (A)

6.2 Period of g (1 mark)

$g(x) = \cos(x - 30^\circ)$; the coefficient of x is 1.

Period = 360° ✓ (A)

6.3 Solve $f(x) = g(x)$ for $x \in [-180^\circ; 180^\circ]$ (5 marks)

$$2 \sin x = \cos(x - 30^\circ)$$

Expand the RHS using $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$: ✓(M)

$$2 \sin x = \cos x \cdot \cos 30^\circ + \sin x \cdot \sin 30^\circ$$

$$2 \sin x = (\sqrt{3}/2) \cos x + (1/2) \sin x \quad \checkmark(A)$$

$$2 \sin x - 1/2 \sin x = (\sqrt{3}/2) \cos x$$

$$(3/2) \sin x = (\sqrt{3}/2) \cos x \quad \checkmark(A)$$

$$\sin x / \cos x = \sqrt{3} / 3$$

$$\tan x = 1/\sqrt{3} = \sqrt{3} / 3 \quad \checkmark(M)$$

Reference angle = 30° . Tan is positive in Q I and Q III.

In $[-180^\circ; 180^\circ]$: $x = 30^\circ$ or $x = 30^\circ - 180^\circ = -150^\circ \quad \checkmark(A)$

$x = -150^\circ$ or $x = 30^\circ$

6.4 Values of x for which $f(x) > g(x)$, $x \in [-180^\circ; 180^\circ]$ (3 marks)

From the graphs and the intersection points found in 6.3: ✓(M)

f is above g between $x = -150^\circ$ and $x = 30^\circ$.

Reading endpoints from the graphs:

- At $x = -150^\circ$, $f = g$ (intersection)
- Between, $f > g$
- At $x = 30^\circ$, $f = g$ (intersection) ✓(A)

$-150^\circ < x < 30^\circ \quad \checkmark(A)$

6.5 Maximum value of $h(x) = f(x) - 1$ (1 mark)

Max $f(x) = 2$; therefore max $h(x) = 2 - 1$.

Maximum of $h = 1 \quad \checkmark(A)$

6.6 Equation of k (g shifted 30° to the LEFT) (2 marks)

Replacing x with $(x + 30^\circ)$ in $g(x) = \cos(x - 30^\circ)$: ✓(M)

$$k(x) = \cos((x + 30^\circ) - 30^\circ) = \cos x$$

$k(x) = \cos x \quad \checkmark(A)$

[13]

QUESTION 7

7.1 Size of angle AFB (1 mark)

In triangle AFB on the ground:

$$\hat{FAB} + \hat{ABF} + \hat{AFB} = 180^\circ \text{ (sum of angles in a triangle)}$$

$$70^\circ + 60^\circ + \hat{AFB} = 180^\circ$$

$$\hat{AFB} = 50^\circ \checkmark (\text{A})$$

7.2 Show that $AF = (50 \sin 60^\circ) / \sin 50^\circ$ (3 marks)

Apply the sine rule in triangle AFB: $\checkmark (\text{M})$

$$AF / \sin(\hat{ABF}) = AB / \sin(\hat{AFB}) \checkmark (\text{R})$$

$$AF / \sin 60^\circ = 50 / \sin 50^\circ \checkmark (\text{A})$$

$$\therefore AF = (50 \sin 60^\circ) / \sin 50^\circ \text{ (as required)}$$

7.3 Height of the tower TF (3 marks)

In right-angled triangle TFA ($TF \perp$ ground):

$$\tan 35^\circ = TF / AF \checkmark (\text{M})$$

$$\therefore TF = AF \cdot \tan 35^\circ$$

$$AF = (50 \sin 60^\circ) / \sin 50^\circ = (50 \times 0,8660) / 0,7660 \approx 56,53 \text{ m} \checkmark (\text{A})$$

$$TF = 56,53 \times \tan 35^\circ = 56,53 \times 0,7002$$

$$TF \approx 39,58 \text{ m} \checkmark (\text{A})$$

7.4 Area of triangle AFB (4 marks)

$$\text{Area} = \frac{1}{2} \times AB \times AF \times \sin(\hat{FAB}) \checkmark (\text{M})$$

$$= \frac{1}{2} \times 50 \times 56,53 \times \sin 70^\circ \checkmark (\text{A})$$

$$= \frac{1}{2} \times 50 \times 56,53 \times 0,9397 \checkmark (\text{M})$$

$$\approx 1\,327,92 \text{ m}^2$$

$$\text{Area of } \triangle AFB \approx 1\,327,92 \text{ m}^2 \checkmark (\text{A})$$

(Alternative: use $\text{Area} = \frac{1}{2} \times AF \times BF \times \sin 50^\circ$ with BF found from sine rule. Both methods give the same result.)

[11]

QUESTION 8

Given: ST is a tangent to the circle at A. ABCD is a cyclic quadrilateral. BD is drawn.
 $\angle TAB = 55^\circ$ and $\angle ABD = 30^\circ$.

8.1 Size of $\angle ADB$ (2 marks)

$\angle TAB = \angle ADB$ (tan-chord angle) ✓(R)

$\therefore \angle ADB = 55^\circ$ ✓(A)

$\angle ADB = 55^\circ$

8.2 Size of $\angle AOB$ (2 marks)

$\angle AOB = 2 \times \angle ADB$ ($\angle$ at centre = $2 \times \angle$ at circumference, same arc AB) ✓(R)

$\therefore \angle AOB = 2 \times 55^\circ = 110^\circ$ ✓(A)

$\angle AOB = 110^\circ$

8.3 Size of $\angle BAD$ (3 marks)

In $\triangle ABD$: $\angle ABD + \angle ADB + \angle BAD = 180^\circ$ (sum of $\angle$ s in a $\triangle$) ✓(R)

$30^\circ + 55^\circ + \angle BAD = 180^\circ$ ✓(M)

$\angle BAD = 180^\circ - 85^\circ$ ✓(A)

$\angle BAD = 95^\circ$

8.4 Size of $\angle BCD$ (2 marks)

$\angle BAD + \angle BCD = 180^\circ$ (opp. $\angle$ s of cyclic quad ABCD) ✓(R)

$95^\circ + \angle BCD = 180^\circ$

$\therefore \angle BCD = 85^\circ$ ✓(A)

$\angle BCD = 85^\circ$

8.5 Size of $\angle AOD$ (2 marks)

$\angle ABD = 30^\circ$ subtends arc AD at the circumference.

$\angle AOD = 2 \times \angle ABD$ ($\angle$ at centre = $2 \times \angle$ at circumference, same arc AD) ✓(R)

$\therefore \angle AOD = 2 \times 30^\circ = 60^\circ$ ✓(A)

$\angle AOD = 60^\circ$

8.6 Show that $\angle SAD = \angle ABD$ (2 marks)

$\angle SAD$ is the angle between the tangent SA and chord AD.

By the tan-chord theorem: $\angle SAD = \text{angle in the alternate segment} = \angle ABD$ ✓(R)

$\therefore \angle SAD = \angle ABD$ (tan-chord) ✓(R)

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QUESTION 9

9.1 Prove that STRU is a parallelogram (4 marks)

To prove: Both pairs of opposite sides are parallel.

$ST \parallel QR$ (given) ✓(R)

Since U lies on QR, the segment UR is part of QR, so $ST \parallel UR$. ✓(M)

$SU \parallel PR$ (given) ✓(R)

Since T lies on PR, the segment TR is part of PR, so $SU \parallel TR$. ✓(M)

∴ STRU has both pairs of opposite sides parallel — it is a parallelogram.

9.2 Ratio ST : QR (3 marks)

In $\triangle PQR$ with $ST \parallel QR$:

$PS / PQ = PT / PR = ST / QR$ (line parallel to one side of a triangle, or $\triangle PST \parallel \triangle PQR$)
✓(R)

Given $PS : SQ = 2 : 3$, so $PS / PQ = 2 / (2 + 3) = 2/5$ ✓(M)

∴ $ST / QR = 2/5$ ✓(A)

ST : QR = 2 : 5

9.3 Ratio SU : PR (3 marks)

Consider $\triangle QPR$. $SU \parallel PR$ with S on PQ and U on QR. ✓(M)

By the same proportionality (or $\triangle QSU \parallel \triangle QPR$): ✓(R)

$QS / QP = SU / PR$

$QS / QP = 3 / (2 + 3) = 3/5$

∴ $SU / PR = 3/5$ ✓(A)

SU : PR = 3 : 5

9.4 Prove that $\triangle PST \parallel \triangle PQR$ (4 marks)

In $\triangle PST$ and $\triangle PQR$:

$\angle PST = \angle PQR$ (corresponding $\angle$ s, $ST \parallel QR$) ✓(R)

$\angle PTS = \angle PRQ$ (corresponding $\angle$ s, $ST \parallel QR$) ✓(R)

$\angle P$ is common. ✓(R)

∴ $\triangle PST \parallel \triangle PQR$ (AAA) ✓(R)

9.5 Ratio area $\triangle PST$: area $\triangle PQR$ (4 marks)

If two triangles are similar, the ratio of their areas equals the square of the ratio of corresponding sides. ✓(R)

From 9.2: $ST / QR = 2/5$ ✓(M)

∴ $\text{area}(\triangle PST) / \text{area}(\triangle PQR) = (2/5)^2$ ✓(M)
 $= 4/25$ ✓(A)

area $\triangle PST$: area $\triangle PQR$ = 4 : 25

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TOTAL: 150